

Group Design Project: Preliminary Design Reports**Assigned: Monday, February 9, 2026****Due: Wednesday, March, 11 2026 via Canvas**

Your current assignment for your Group Design Project is to design a Rankine power cycle (see lecture notes and handout with design table, posted to Canvas) based on your assigned sustainable/renewable energy source to produce the electrical power specified in/based on your Proposals.

Common Questions to Be Addressed in All Designs

- 1) Specify your electrical power to be supplied as the net power for your cycle ($\dot{W}_{net}$, in kW, MW, or GW). Assume a constant power demand that does not vary with time. See the specific comments for your group in your evaluated submission for your proposal and in the document entitled “Group Design Project: Specific Guidance for Individual Groups for Preliminary Design Reports” (to be posted on Canvas after proposal submission).
- 2) Complete a design table for an ideal Rankine cycle (available in the handout entitled Process Flow Diagram (PFD) & Table for Designing a Rankine Power Cycle”, posted to Canvas) based on your assigned sustainable/renewable energy source, identifying the properties for each of the four states in the cycle.
- 3) Specify the required cooling as the net cooling (heat removal) for your data center (Q in kW, MW, or GW).
- 4) Complete a design table for a vapor-compression cooling cycle (available in the handout entitled Process Flow Diagram (PFD) & Table for Designing a Vapor-Compression Refrigeration Cycle”, posted to Canvas) based on either liquid or air cooling, identifying the properties for each of the four states in the cycle.
- 5) Determine for ideal cycles of power and refrigeration (*i.e.*, assuming isentropic efficiencies are 100% for both the turbine & compressor) based on your assigned sustainable/renewable energy source and cooling approach:
 - a) The mass flow rate for the working fluids for the cycle, $\dot{m}$ (in kg/s).
 - b) The required heat transfer rate from the hot reservoir to the working fluids, $\dot{Q}_H$ (in same units as power).
 - For biomass also specify the mass rate (in kg/s) at which your fuel is burned.
 - For geothermal also specify the mass flow rate (in kg/s) required from your geothermal source.
 - For ocean thermal also specify the mass flow rate (in kg/s) at which warm water is supplied to boiler.
 - For solar thermal also specify the size (in both m² and football fields, where one football field has an area of approximately 5350 m²) for the solar collector.
 - c) The required heat transfer rate to the hot/cold reservoir from the working fluid, in same units as power), and the required mass flow rate (in kg/s) of cooling water.
 - d) The efficiency for the cycles, η .
- 6) Determine the specified properties for your process listed Question 3 for a corresponding non-ideal cycle with an isentropic efficiency of 75% for any turbine and 70% for any compressor based on your assigned sustainable/renewable energy source. Compare your results for ideal and non-ideal cycles for the properties you are asked to report using a table. Comment on whether your results make sense.

Note that you do not need to find an existing design for your cycles in the literature!

Guidelines for Submission

- 1) Each team should submit a single PDF with their work, including a design table.
- 2) Include sufficient details for the reader to easily verify your outcomes.
- 3) Scanned handwritten work with legible script is permissible.
- 4) Include a cover page with your project name, the names of each team member alphabetized based on last name, the course name, and date of submission.
- 5) There is no page limit. Use 1" margins on all sides, and place a page number at the bottom of each page after the cover page.

Please review the rubric with detailed breakdown of criteria for assessment below before submitting your preliminary design report to confirm key elements sought. Note the following:

- You will have the opportunity to correct errors as part of your final design report.

As always, groups are welcome and encouraged to discuss their projects, their ideas, and problems encountered with the instructor prior to and after submission of preliminary design reports. Keep in mind that earlier discussions often are more fruitful (and pleasant) than last-minute efforts!

General Guidance for All Groups

The inlet to the turbine may be either a superheated vapor or a supercritical fluid.

Specific Guidance for Individual Groups

See accompanying document entitled “Group Design Project: Specific Guidance for Individual Groups for Preliminary Design Reports” (posted to Canvas along with proposal feedback) for specific design metrics for your group. “Group choice” means that you may choose a value consistent with your situation.

Specific Guidance for Alternative Strategies

Use of fluids other than water as working fluids

- Tabulated data for these fluids is available through the NIST database.

Use of cooling water for condenser

- You may assume a constant specific heat capacity at constant pressure for water and the temperature rise must be no more than 2 °C).
- You will need an estimate for the temperature for the source of your cooling water; this temperature will guide your selection of T_2 for your cycle.

Use of biomass/biofuel to heat boiler

- You will need to identify your biomass source or biofuel and/or an estimate for the fuel value ($\Delta \hat{h}_{comb}$, in units of energy per unit mass) for combustion for your fuel if you have not identified one previously.
- Apply the relationship $\dot{Q}_H = \dot{m}_{fuel} \Delta \hat{h}_{comb}$, where $\dot{m}_{fuel}$ is the mass rate at which fuel is

burned and $\Delta \hat{h}_{comb}$ the fuel value (in units of energy/unit mass) for its combustion.

- Discuss limits on the temperature for inlet to turbine based on your biomass/biofuel.

Use of geothermal water for boiler for geothermal processes

- You will need to estimate the state (*i.e.*, phase(s), temperature, and pressure) for your geothermal source if you have not identified this state previously.
- Identify how you chose the turbine inlet temperature based on your geothermal source temperature.
- Assume the wastewater coming out of the geothermal side of the boiler is injected back into the ground, albeit at a lower specific enthalpy than it is pumped from the geothermal source and a temperature greater than the inlet to the turbine.
- Apply the relationship $\dot{Q}_H = \dot{m}_{\text{geothermal}} (\hat{h}_{\text{geothermalsource}} - \hat{h}_{\text{injection}})$, where $\dot{m}_{\text{geothermal}}$ is the mass flow rate and $\hat{h}_{\text{geothermalsource}}$ the specific enthalpy, respectively, of your geothermal source and $\hat{h}_{\text{injection}}$ is the specific enthalpy of the injected wastewater.
- Note that Wolfram Alpha provides a convenient platform for estimating the specific enthalpies of subcooled liquids at pressures below what Koretsky provides.

Use of water for ocean thermal processes

- Consider only a closed process based on a Rankine cycle for your preliminary design.
- You will need estimates for temperatures for the hot feed (from at or near the ocean surface) and cold feed (at your specified depth) for your location if you have not identified both of these temperatures previously.
- Identify how you chose the turbine inlet temperature based on your hot feed temperature.
- Apply the relationship $\dot{Q}_i = \dot{m}_i \hat{c}_p \Delta T$, where $\dot{m}_i$ is the mass flow rate of water stream i and $\hat{c}_p = 4.184 \text{ J/g} \cdot \text{K}$.

Heat transfer from solar thermal collectors

- Identify how you chose the turbine inlet temperature based on your collector temperature.
- Apply the relationship $\dot{Q}_H = r A_{\text{collector}} j$, where $A_{\text{collector}}$ & r are the area and concentration ratio, respectively, of your collector and j is the net radiative energy flux from the sun.
- You will need to identify your type of collector and/or an estimate for both the concentration ratio and the temperature for your collector if you have not done so previously..
- Smil (2010) suggests net radiative energy fluxes of 230 W/m^2 and 100 W/m^2 are representative for sunny desert and cloudy northern locations, respectively. For your specific projects use $j = 230 \text{ W/m}^2$ for locations in Hawaii, the southwestern US, and the Middle East, $j = 200 \text{ W/m}^2$ for locations in the southeastern US or northern California, and $j = 170 \text{ W/m}^2$ for locations in the Midwest or Northeast (based on data from <http://www.pveducation.org/pvcdrom/properties-of-sunlight/average-solar-radiation>).

Additional Guidance on Designing Non-ideal Rankine Cycles

The following strategy enables the design of a Rankine cycle operated irreversibly (*i.e.*, a cycle for which the isentropic efficiencies of the turbine and/or compressor are not 100%):

- 1) Complete the design table for an ideal Rankine cycle (*i.e.*, a reversible cycle for which the turbine & compressor are operated isentropically).

- 2) Apply the strategy introduced in the Handout for Lecture #12: Practice Problem on Comparing the Designs for Ideal vs. Non-ideal Rankine Cycles to relate the actual mass flow rate, $\dot{m}_{actual}$, to the actual net power produced, $\dot{W}_{net,actual}$.

a) Define the isentropic efficiencies for the turbine & compressor respectively as:

$$\eta_{turbine} \equiv \frac{\dot{W}_{s,turbine,actual}}{\dot{W}_{s,turbine,rev}} = \frac{\dot{m}_{actual}(\hat{h}_{2,act} - \hat{h}_{1,act})}{\dot{m}_{actual}(\hat{h}_{2,rev} - \hat{h}_{1,rev})} = \frac{\hat{h}_{2,act} - \hat{h}_{1,act}}{\hat{h}_{2,rev} - \hat{h}_{1,rev}}$$

$$\eta_{compressor} \equiv \frac{\dot{W}_{s,compressor,rev}}{\dot{W}_{s,compressor,actual}} = \frac{\dot{m}_{actual}(\hat{h}_{4,rev} - \hat{h}_{3,rev})}{\dot{m}_{actual}(\hat{h}_{4,act} - \hat{h}_{3,act})} = \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\hat{h}_{4,act} - \hat{h}_{3,act}}$$

- b) Rearrange these expressions to express the actual power produced by the turbine and consumed by the compressor, respectively, in terms of the actual mass flow rate as:

$$\dot{W}_{s,turbine,actual} = \eta_{turbine} \dot{W}_{s,turbine,rev} = \eta_{turbine} \dot{m}_{actual} (\hat{h}_{2,rev} - \hat{h}_{1,rev})$$

$$\dot{W}_{s,compressor,actual} = \frac{\dot{W}_{s,compressor,rev}}{\eta_{compressor}} = \frac{\dot{m}_{actual} (\hat{h}_{4,rev} - \hat{h}_{3,rev})}{\eta_{compressor}}$$

- c) The actual net power is the sum of the actual power produced by the turbine and actual power consumed by the compressor:

$$\dot{W}_{net,actual} = \dot{W}_{s,turbine,actual} + \dot{W}_{s,compressor,actual}$$

$$= \eta_{turbine} \dot{m}_{actual} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\dot{m}_{actual} (\hat{h}_{4,rev} - \hat{h}_{3,rev})}{\eta_{compressor}}$$

$$\dot{W}_{net,actual} = \dot{m}_{actual} \left[\eta_{turbine} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \right]$$

- d) For your designs the actual net power should be the same as the net power for the reversible cycle as you are specifying the net power to be supplied. Rearrange to solve for the actual mass flow rate for the working fluid:

$$\dot{m}_{actual} = \frac{\dot{W}_{net,actual}}{\eta_{turbine}(\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}}}$$

- 3) To determine the actual rates of heat exchange into the working fluid in the boiler, $\dot{Q}_{H,actual}$, and out of the working fluid in the condenser, $\dot{Q}_{C,actual}$, requires recognizing that irreversible operation means that not all of the specific enthalpies identified for the ideal (reversible) cycle are applicable for the non-ideal cycle. In particular, define the actual rates of heat transfer as respectively:

$$\dot{Q}_{H,actual} = \dot{m}_{actual}(\hat{h}_{1,act} - \hat{h}_{4,act}) \quad \dot{Q}_{C,actual} = \dot{m}_{actual}(\hat{h}_{3,act} - \hat{h}_{2,act})$$

Note that the differences between the actual specific enthalpies for States 1 & 2 and between the actual specific enthalpies for States 3 & 4 are related through the relationships for isentropic efficiency identified for the turbine & compressor above:

$$\hat{h}_{2,act} - \hat{h}_{1,act} = \eta_{turbine}(\hat{h}_{2,rev} - \hat{h}_{1,rev}) \quad \hat{h}_{4,act} - \hat{h}_{3,act} = \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}}$$

The above pair of relationships replaces the isentropic relationships $\hat{s}_1 = \hat{s}_2$ & $\hat{s}_3 = \hat{s}_4$ in constraining the operation of our cycle and indicate that we can set independently either State 1 or State 2 and either State 3 or State 4.

To determine which pair of states to specify we need to consider which operating constraint is higher priority: the temperature for the working fluid flowing through the condenser (constraining States 2 & 3) or through the boiler (constraining States 4 & 1). Given that for all groups the temperature of the cooling water imposes an immediately-identifiable constraint, we first consider fixing T_2 for the non-ideal cycle to the value for the ideal cycle (*i.e.*, the properties for States 2 & 3 for the ideal cycle apply also for the non-ideal cycle, such that $\hat{h}_{2,act} = \hat{h}_{2,rev}$ & $\hat{h}_{3,act} = \hat{h}_{3,rev}$ with $x_{2,act} = x_{2,rev}$). We then find:

$$\begin{aligned} \hat{h}_{1,act} &= \hat{h}_{2,act} - \eta_{turbine}(\hat{h}_{2,rev} - \hat{h}_{1,rev}) = \hat{h}_{2,rev} - \eta_{turbine}(\hat{h}_{2,rev} - \hat{h}_{1,rev}) \\ \hat{h}_{4,act} &= \hat{h}_{3,act} + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} = \hat{h}_{3,rev} + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \end{aligned}$$

such that:

$$\begin{aligned}\dot{Q}_{H,actual} &= \dot{m}_{actual} \left[\left[\hat{h}_{2,rev} - \eta_{turbine} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) \right] - \left[\hat{h}_{3,rev} + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \right] \right] \\ \dot{Q}_{H,actual} &= \dot{m}_{actual} (\hat{h}_{2,rev} - \hat{h}_{3,rev}) - \dot{m}_{actual} \left[\eta_{turbine} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \right] \\ \dot{Q}_{C,actual} &= \dot{m}_{actual} (\hat{h}_{3,rev} - \hat{h}_{2,rev})\end{aligned}$$

We verify that the overall energy balance for the cycle is satisfied:

$$\begin{aligned}\frac{dU_{cycle}}{dt} &= \dot{W}_{net,actual} + \dot{Q}_{H,actual} + \dot{Q}_{C,actual} \\ &= \dot{m}_{actual} \left[\eta_{turbine} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \right] \\ &\quad + \dot{m}_{actual} (\hat{h}_{2,rev} - \hat{h}_{3,rev}) - \dot{m}_{actual} \left[\eta_{turbine} (\hat{h}_{2,rev} - \hat{h}_{1,rev}) + \frac{\hat{h}_{4,rev} - \hat{h}_{3,rev}}{\eta_{compressor}} \right] \\ &\quad + \dot{m}_{actual} (\hat{h}_{3,rev} - \hat{h}_{2,rev}) \\ \frac{dU_{cycle}}{dt} &= 0\end{aligned}$$

It is critical that we now confirm that our specifications for operation of the boiler are satisfied (*i.e.*, the temperatures for States 1 & 4 must be less than the temperature of the hot reservoir). Note that, alternatively, we could consider fixing T_1 & T_4 for the non-ideal cycle to the values for the ideal cycle (*i.e.*, the properties for States 1 & 4 for the ideal cycle apply also for the non-ideal cycle, such that $\hat{h}_{1,act} = \hat{h}_{1,rev}$ & $\hat{h}_{4,act} = \hat{h}_{4,rev}$). We leave further development of this consideration to the interested reader.

Rubric for AssessmentPart 1—Specifying Power to Be Supplied:Specified explicitly $\dot{W}_{net} < 0$

Specified power appropriately

Part 2—Design Tables Based on Ideal Cycle:

Provided design table

Design table matches numbering scheme for states on PFD

Completed design table for each state for:

P	T	$\hat{h}$ or h	$\hat{s}$ or s	x_2 (or $f_{v,2}$)
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Turbine is isentropic

Compressor is isentropic

Identified explicitly how temperature for inlet for turbine was chosen

Correctly satisfied constraints on State 1

Identified appropriately inlet temperature of cooling water

Correctly satisfied constraints on State 2

Part 3—Design for Ideal Cycles:

Determined mass flow rate for working fluid

Determined $\dot{Q}_H$ for cycleDetermined $\dot{Q}_C$ for cycle

Determined specifications for energy source

Determined mass flow rate for cooling water

Determined overall efficiency for cycle

Part 4—Design for Non-Ideal Cycles:

Determined mass flow rate for working fluid

Determined $\dot{Q}_H$ for cycleDetermined $\dot{Q}_C$ for cycle

Determined specifications for energy source

Determined mass flow rate for cooling water

Determined overall efficiency for cycle

Commented appropriately whether results for non-ideal cycle make sense compared to results for ideal cycle, including comparing rates of heat transfer in boiler and condenser for ideal vs. non-ideal cycle

Assessment: Scores of Meets Expectations will be earned with no more than 2 moderate deficiencies, with all work being easy to follow/understand. Scores of Excellent can be achieved with no more than two minor deficiencies and a polished quality of work.

A minor deficiency for example might be a math/calculator error or misreading a number from a table. A moderate deficiency for example might be a value that is missing or more significant math errors involving mishandling of units, etc.